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GEOGRAPHY · India: physiographic divisions

UNDERSTAND IT FIRST

What is it? Physiography means the **shape of the land**: mountains, plains, plateaus, deserts, coasts and islands. India's relief was made over very long periods: the **Peninsular Plateau** is one of the **oldest landmasses** (part of **Gondwana**), the **Himalayas** are **young fold mountains** pushed up when the Indian plate hit Asia, and the **Northern Plains** were filled in by river **silt**. Each division has its own **soils, crops, rivers and people**. Learn it as **India's size and location → Himalayas → Northern Plains → Peninsular Plateau → Desert → Coastal Plains → Islands → AP**.

1. Size and location

Area about **32.87 lakh sq km**, about **2.4%** of the world's land: the **7th largest** country. Mainland latitudes **8°4′ N to 37°6′ N**; longitudes **68°7′ E to 97°25′ E**. North–south about **3,214 km**; east–west about **2,933 km**. Land frontier about **15,200 km**; the coastline was long given as **7,516.6 km** (with islands), and a **2025 re-measurement** put it at about **11,098 km** (check the latest). The **Tropic of Cancer** divides India almost in half; the **standard meridian** is **82°30′ E**. India's **southernmost point** is **Indira Point** (Great Nicobar).

Six physiographic divisions

2. The Himalayas

Cross-section of the Himalayas

Range	Features
Greater Himalaya (Himadri)	The **highest and most continuous** range, average about **6,000 m**, with a **granite core** and permanent snow; peaks: **Everest (8,849 m, Nepal)**, **Kangchenjunga (8,586 m, highest in India proper, Sikkim)**, Nanga Parbat, Nanda Devi; passes: **Zoji La, Shipki La, Nathu La**.
Lesser Himalaya (Himachal)	**3,700–4,500 m**; ranges **Pir Panjal, Dhauladhar, Mahabharat**; valleys of **Kashmir, Kullu, Kangra**; hill stations **Shimla, Mussoorie, Nainital, Darjeeling**.
Shiwaliks (Outer Himalaya)	The **outermost and lowest** range (**900–1,100 m**), made of **loose river sediments**; long flat valleys between them and the Lesser Himalaya are **duns** (**Dehra Dun**, Kotli Dun, Patli Dun).
Trans-Himalaya and Purvanchal	North of the Greater Himalaya: **Karakoram** (**K2, 8,611 m**, India's claimed highest), **Ladakh, Zaskar**. In the east, the Himalaya bends south as the **Purvanchal**: **Patkai Bum, Naga hills, Manipur hills, Mizo (Lushai) hills**.

Regional divisions (west to east)	**Punjab Himalaya** (Indus–Sutlej), **Kumaon Himalaya** (Sutlej–Kali), **Nepal Himalaya** (Kali–Teesta), **Assam Himalaya** (Teesta–Dihang).

3. The Northern Plains

Belts of the Northern Plains

Formed by the **Indus, Ganga and Brahmaputra** and their tributaries; about **2,400 km long**, among the **most fertile and densely populated** regions in the world. Three parts: the **Punjab plains** (land between two rivers is a **doab**), the **Ganga plains** and the **Brahmaputra plains** (with **Majuli**, among the world's largest river islands). Saline patches in dry areas are called **reh or kallar**.

4. The Peninsular Plateau and the Desert

- **Peninsular Plateau:** a roughly **triangular** block of **old, hard rocks**. **Central Highlands** (north of the Narmada): **Malwa plateau**, **Bundelkhand, Baghelkhand** and the mineral-rich **Chota Nagpur**; ranges: **Aravallis** (old, worn; highest **Guru Shikhar, 1,722 m**, Mount Abu), **Vindhyas** and **Satpura**. **Deccan Plateau** (south of the Narmada), sloping **east**, so most rivers flow **east into the Bay of Bengal**; its north-west is covered by the **Deccan Traps** (lava). The **Meghalaya (Shillong) plateau** (Garo, Khasi, Jaintia hills) is part of the peninsula, cut off by the **Malda gap**.
- **Western and Eastern Ghats:** the **Western Ghats (Sahyadri)** are **higher and continuous**, crossed only by passes (**Thal, Bhor, Pal ghats**); highest peak **Anamudi (2,695 m)**, then **Doddabetta (2,637 m)** in the **Nilgiris**, where the two Ghats meet. The **Eastern Ghats** are **lower and broken** by the **Mahanadi, Godavari, Krishna and Kaveri**; high points include **Mahendragiri** (Odisha) and **Arma Konda** (about **1,680 m**, AP, the highest).
- **Indian (Thar) Desert:** west of the **Aravallis**; rainfall usually **below 150 mm**; **sand dunes (barchans)**; the **Luni** is the only large river; salt lakes such as **Sambhar**. The Aravallis lie **parallel** to the south-west monsoon winds, so they do not stop them to cause rain.

5. Coastal Plains and Islands

Part	Features
Western coastal plain	**Narrow**; from north: **Kachchh and Kathiawar**, **Konkan** (Mumbai–Goa), **Kannad** (Goa–Mangaluru), **Malabar** (Kerala) with **backwaters (kayals)** such as **Vembanad**; west-flowing rivers form **estuaries**, not deltas.
Eastern coastal plain	**Wider**, with big **deltas** (**Mahanadi, Godavari, Krishna, Kaveri**); the **Northern Circars** (Mahanadi to Krishna, including most of the **AP coast**) and the **Coromandel coast** (Krishna to Kanyakumari); lagoons **Chilika** (largest brackish lagoon) and **Pulicat**; **Kolleru** lake between the Godavari and Krishna deltas.
Andaman and Nicobar	Over **500** islands in the Bay of Bengal, thought to be the **raised parts of submerged mountain ranges**; the **Ten Degree Channel** separates the Andamans from the Nicobars; highest point **Saddle Peak**; **Barren Island** (active volcano) and **Narcondam** (dormant).

Lakshadweep	**36 coral islands** in the Arabian Sea (atolls); capital **Kavaratti**; the **Nine Degree Channel** separates **Minicoy** from the main group, and the **Eight Degree Channel** separates Minicoy from the **Maldives**. The **Palk Strait** lies between India and **Sri Lanka**.

Andhra Pradesh: three relief zones: the **coastal plains and deltas** (Krishna–Godavari, Kolleru, Pulicat); the **Eastern Ghats** (**Araku and Ananthagiri** hills in the north, **Papikonda** at the Godavari gorge, **Nallamala, Velikonda, Erramala, Seshachalam and Palakonda** in the south, and **Horsley Hills** near Madanapalle); and the **Rayalaseema plateau**, part of the Deccan.

EXAM TRAP

Area **32.87 lakh sq km**, **2.4%**, **7th**; **8°4′–37°6′ N**, **68°7′–97°25′ E**; **3,214 km** N–S, **2,933 km** E–W; land border **~15,200 km**; coast **7,516.6 km** (re-measured **~11,098 km**, 2025); **82°30′ E**; **Indira Point**; **Himadri** (highest, continuous, **~6,000 m**; Everest **8,849**, Kangchenjunga **8,586**; Zoji La, Shipki La, Nathu La); **Himachal** (Pir Panjal, Dhauladhar; Shimla, Darjeeling); **Shiwaliks** (outermost, **duns**); Trans (**Karakoram, K2 8,611**); **Purvanchal** (Patkai, Naga, Mizo); Punjab, Kumaon, Nepal, Assam Himalaya; plains: **bhabar, terai, bhangar (old), khadar (new)**; doab; **Majuli**; plateau: **Malwa, Bundelkhand, Chota Nagpur**; **Aravallis (Guru Shikhar 1,722)**; Deccan slopes **east**; **Deccan Traps**; **Meghalaya plateau, Malda gap**; **Western Ghats continuous (Anamudi 2,695; Doddabetta 2,637)**; **Eastern Ghats broken (Arma Konda ~1,680)**; **Thar** (<150 mm; Luni; Sambhar); west coast **narrow** (Kachchh, Konkan, Kannad, Malabar; kayals); east coast **wide** (deltas; **Northern Circars, Coromandel**; Chilika, Pulicat, Kolleru); **A&N (Ten Degree Channel; Saddle Peak; Barren, Narcondam)**; **Lakshadweep (36, coral, Kavaratti; 9° and 8° Channels)**; **Palk Strait**; AP: coast, **Eastern Ghats (Araku to Seshachalam, Horsley Hills)**, Rayalaseema plateau.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:
1. The Greater Himalaya (Himadri) is the highest and most continuous range.
2. The Shiwaliks are the outermost range of the Himalayas.
3. The bhabar belt lies to the south of the terai.
4. Khadar is made of newer alluvium renewed by floods.
Which of the above are correct?

A. 1, 2 and 4 only

B. 1 and 2 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. The bhabar lies at the foot of the Shiwaliks, north of the terai.

PRACTICE

Match List-I (water body) with List-II (what it separates):

List-I (water body)	List-II (what it separates)
A. Ten Degree Channel	1. India and Sri Lanka
B. Nine Degree Channel	2. Minicoy and the Maldives
C. Eight Degree Channel	3. Minicoy and the main Lakshadweep group
D. Palk Strait	4. The Andaman and the Nicobar Islands

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 2 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (3) 10°: Andaman–Nicobar; 9°: Minicoy–Lakshadweep; 8°: Minicoy–Maldives; Palk Strait: India–Sri Lanka.

PRACTICE

Which one of the following statements is NOT correct?

A. The Western Ghats are more continuous than the Eastern Ghats.

B. Anamudi is the highest peak in south India.

C. Arma Konda in Andhra Pradesh is the highest point of the Eastern Ghats.

D. The eastern coastal plain is narrower than the western coastal plain.

Answer: D. The eastern coastal plain is wider, with large deltas; the western one is narrow.

PRACTICE

Assertion (A): The Lakshadweep islands are of coral origin.
Reason (R): Barren Island is India's only active volcano.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true, but Barren Island (in the Andaman Sea) has nothing to do with the coral origin of Lakshadweep.

PRACTICE

A traveller drives along the coast from the Mahanadi delta south to the Krishna delta. Which coastal plain is this?

A. Konkan

B. Malabar

C. Northern Circars

D. Coromandel

Answer: C. The Northern Circars run from the Mahanadi to the Krishna; the Coromandel coast runs from the Krishna to Kanyakumari.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write India's size and location figures. 2. Write the Himalayan ranges with heights, peaks, passes and regional divisions, and the belts of the plains. 3. Write the plateau parts, the two Ghats with peaks, the desert, the coasts, the islands with channels, and AP's relief.

ECONOMY · The New Economic Reforms of 1991

UNDERSTAND IT FIRST

What is it? In **1991** India faced a severe **balance-of-payments crisis**: it had almost **no foreign exchange** left to pay for imports. The Government of **P.V. Narasimha Rao**, with **Finance Minister Manmohan Singh**, responded with sweeping **reforms**, known as **LPG: Liberalisation, Privatisation and Globalisation**. They ended the **'licence raj'**, opened India to **foreign trade and investment**, and shrank the State's direct role in business. They changed India's growth path. Learn it as **why the crisis came → immediate rescue → stabilisation vs structural reform → LPG measures → committees → results and criticisms → later reforms**.

1. The crisis and the rescue

- **Causes:** large **fiscal deficits** through the 1980s (about **8% of GDP** for the Centre); growing **foreign debt**; the **Gulf War (1990–91)** pushed up **oil prices** and cut **remittances** from Indian workers in the Gulf; **political instability**; **inflation** in double digits; a **credit-rating downgrade**, after which **NRIs withdrew deposits**. By mid-1991, **foreign exchange reserves** had fallen to about **$1 billion**, enough for only about **two weeks of imports**.
- **Rescue:** India **pledged its gold** (about **47 tonnes** with the **Bank of England and Bank of Japan** in July 1991, after an earlier sale-and-repurchase of 20 tonnes) to raise dollars, and took loans from the **IMF** (with conditions) and the **World Bank**. The **rupee was devalued** in two steps (**1 and 3 July 1991**, about **18–19%**). The **New Industrial Policy** and **Union Budget** were presented on **24 July 1991**.
- **Two kinds of measures: stabilisation** (short-term, linked to the **IMF**): cut the **fiscal deficit**, control **inflation** and fix the **balance of payments** by managing **demand**. **Structural adjustment** (long-term, linked to the **World Bank**): change the **supply side** of the economy through **LPG**, so it grows faster and competes globally.

2. The LPG measures

Area	What changed
Industry	**Industrial licensing abolished** for all but a few industries (now only a handful, such as **defence, hazardous chemicals, tobacco and alcohol**); industries **reserved for the public sector** cut from **17 to 8**, and now only **2** (**atomic energy** and **railway operations**); **MRTP** limits on big firms removed (later replaced by the **Competition Act, 2002**; **CCI**).
Trade and foreign exchange	**Import licensing** relaxed and **customs duties cut** sharply; a **market-determined exchange rate** (1993); **current account convertibility** (**1994**); **FERA** replaced by the milder **FEMA (1999)**; India joined the **WTO** as a founding member (**1995**). **Full capital account convertibility** has **not** been adopted (Tarapore committees, 1997 and 2006).
Foreign investment	**FDI** allowed up to **51%** automatically in many industries (now up to 100% in most sectors); **foreign institutional investors** allowed into stock markets; **SEBI** became a statutory regulator (**1992**).
Privatisation	**Disinvestment** of government shares in public sector units; the **Department of Disinvestment** later became **DIPAM (2016)**. **Strategic disinvestment** means selling **management control**, e.g. **Air India to the Tata group (January 2022)**; the **LIC IPO** came in **2022**.
Committees	**Narasimham Committee** (**1991**, **1998**): **banking** reforms (lower CRR and SLR, new private banks, prudential norms); **Raja Chelliah Committee (1991)**: **tax** reforms; **Malhotra Committee (1993)**: **insurance** reforms, leading to **IRDA (1999)**; **Rangarajan Committee (1993)**: **disinvestment**.

3. Results, criticisms and later reforms

- **Gains:** faster growth (about **6–7%** on average, compared with the 'Hindu rate'); **forex reserves** rose to **over $600 billion** in recent years; booms in **IT, telecom, banking and services**; more consumer choice; India became a major destination for **foreign investment**.
- **Criticisms: agriculture** got less public investment; **jobless growth** and **informal** work; rising **inequality**; small industries hurt by **cheap imports**; services grew but **manufacturing jobs** lagged; reforms were sometimes called **'reforms by stealth'** because they came without wide public debate.
- **Second-generation reforms: FRBM Act (2003)** for fiscal discipline; the **Insolvency and Bankruptcy Code (2016)**; **inflation targeting** with a **Monetary Policy Committee (2016)**; **GST (2017)**; the **four Labour Codes** (in force **2025**); **Make in India (2014)** and **PLI** schemes for manufacturing.

EXAM TRAP

Crisis 1991: fiscal deficit **~8%**, debt, **Gulf War** (oil, remittances), instability, inflation, rating cut, NRIs withdrew; reserves **~$1 bn (~2 weeks of imports)**; **gold pledged (~47 t; BoE, BoJ)**; **IMF, World Bank**; **devaluation 1 and 3 July 1991 (~18–19%)**; **NIP and Budget 24 July 1991**; **P.V. Narasimha Rao, Manmohan Singh**; **stabilisation (IMF, short, demand)** vs **structural adjustment (World Bank, long, supply)**; licensing abolished (few left); public sector reserved **17 → 8 → 2** (atomic energy, railway operations); **MRTP → Competition Act 2002 (CCI)**; **FERA → FEMA 1999**; market exchange rate **1993**; **current account convertibility 1994**; no full capital account convertibility (**Tarapore**); **WTO 1995**; FDI **51%** automatic; **SEBI statutory 1992**; **DIPAM 2016**; **Air India–Tata Jan 2022**; **LIC IPO 2022**; **Narasimham (1991, 1998) banking**; **Chelliah tax**; **Malhotra insurance → IRDA 1999**; **Rangarajan disinvestment 1993**; gains (growth, reserves **$600 bn+**, IT); criticisms (agriculture, jobless, inequality); **FRBM 2003, IBC 2016, MPC 2016, GST 2017, Labour Codes 2025**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the 1991 crisis:
1. India's foreign exchange reserves fell to barely enough for about two weeks of imports.
2. India pledged gold to raise foreign currency.
3. The Gulf War raised India's oil import bill.
4. India had a large current account surplus at the time.
Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2, 3 and 4

D. 1, 2 and 3 only

Answer: D. India had a large current account deficit, which is why reserves ran out.

PRACTICE

Match List-I (committee) with List-II (area of reform):

List-I (committee)	List-II (area of reform)
A. Narasimham Committee	1. Disinvestment of public sector units
B. Raja Chelliah Committee	2. Insurance sector
C. Malhotra Committee	3. Tax system
D. Rangarajan Committee (1993)	4. Banking sector

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 3 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (2) Narasimham: banking; Chelliah: tax; Malhotra: insurance; Rangarajan: disinvestment.

PRACTICE

Which one of the following statements is NOT correct?

A. India introduced full capital account convertibility in 1991.

B. The rupee was devalued in July 1991.

C. India adopted current account convertibility in 1994.

D. India was a founding member of the WTO in 1995.

Answer: A. India has only current account convertibility; full capital account convertibility has not been adopted.

PRACTICE

Assertion (A): The New Industrial Policy of 1991 abolished industrial licensing for most industries.
Reason (R): The policy increased the number of industries reserved for the public sector.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true. R is false: the reserved list was cut from 17 to 8 (now 2).

PRACTICE

The Government sells a majority stake and full management control of a loss-making public sector airline to a private group. What is this called?

A. Nationalisation

B. Strategic disinvestment

C. Minority stake sale

D. Merger

Answer: B. Selling control along with shares is strategic disinvestment, as in Air India's sale to the Tata group (2022).

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the causes of the 1991 crisis with figures and the rescue steps with dates. 2. Write stabilisation vs structural adjustment and the LPG measures by area. 3. Write the committees, results, criticisms and second-generation reforms.

SCIENCE · Pollution control: laws, regulation and recent initiatives

UNDERSTAND IT FIRST

What is it? Day 30 explained **what pollution is and what it does**. Today is about **how India controls it**: **laws** passed by Parliament, **rules and notifications** made under them, **regulators** (the Pollution Control Boards and the NGT), **principles laid down by the courts**, **international agreements**, and **new schemes** for clean air, clean rivers and clean transport. Learn it as **laws in time order → tools of regulation (consent, EIA, CRZ, industry categories) → NGT → court principles → Montreal and Kigali → recent initiatives → AP**.

India's main pollution-control laws

1. Laws and regulation

Law or tool	Key points
Water Act, 1974	India's **first major pollution law**; set up the **Central and State Pollution Control Boards**; industries need the Board's **'consent to establish'** and **'consent to operate'**. A **2024 amendment** replaced jail terms for minor violations with **penalties**.
Air Act, 1981	Passed after the **UN Stockholm Conference (1972)**; States can declare **air pollution control areas**; the same Boards enforce it.
Environment (Protection) Act, 1986	An **umbrella law** passed after the **Bhopal gas tragedy**; gives the Centre wide powers to set **standards** and make rules. Under it come the **EIA Notification (1994, revised 2006)**: big projects need **environmental clearance** after an impact study and a **public hearing** (**Category A** by the Centre, **Category B** by State authorities); the **Coastal Regulation Zone (CRZ) Notification (1991; 2011; 2019)**; and all the **waste rules**. The **Jan Vishwas Act (2023)** replaced jail with **penalties** for many offences under this Act and the Air Act.
Industry categories	Since **2016**, industries are grouped by pollution potential: **Red** (most polluting), **Orange**, **Green**, and **White** (practically non-polluting; no consent needed). The most polluting units must install **continuous emission and effluent monitoring** linked online to the Boards.
National Green Tribunal (2010)	A specialised court for **environmental cases**; principal bench in **New Delhi**, with zonal benches (**Bhopal, Pune, Kolkata, Chennai**); chaired by a **retired Supreme Court judge** or **retired High Court Chief Justice**; aims to decide cases within **six months**; applies **sustainable development, precautionary and polluter-pays** principles; appeals go to the **Supreme Court**.

2. Principles from the courts and international agreements

Principle	Case
Absolute liability: a hazardous industry must pay for all harm, with no excuses	**M.C. Mehta (Oleum gas leak), 1986**
Right to a pollution-free environment is part of the **right to life (Art 21)**	**Subhash Kumar v. State of Bihar, 1991**
Polluter pays and **precautionary principle** are part of Indian law	**Vellore Citizens' Welfare Forum, 1996**
Public trust doctrine: the State holds rivers, forests and the sea in trust for the people	**M.C. Mehta v. Kamal Nath, 1997**

Ozone layer: the **Vienna Convention (1985)** and the **Montreal Protocol (1987)** phase out **ozone-depleting substances** such as **CFCs**; it is seen as the most successful environmental treaty. The **Kigali Amendment (2016)** adds a **phase-down of HFCs**, which do not harm ozone but are **powerful greenhouse gases**; India will freeze HFC use in **2028** and cut it from **2032**. India's **Cooling Action Plan (2019)** supports this.

3. Recent initiatives

- **Clean air: NCAP (2019)** with the **PRANA portal** to track city action plans; **Swachh Vayu Survekshan** (since **2023**) ranks cities on clean-air work; **CAQM (2021)** and **GRAP** for Delhi–NCR; **Surat** launched the world's **first emissions trading scheme for particulate matter (2019)**.
- **Clean transport and fuel: BS-VI (2020)**; **FAME I (2015)** and **FAME II (2019)** for electric vehicles, followed by **PM E-DRIVE (2024)** and **PM-eBus Sewa (2023)**; the **Vehicle Scrappage Policy (2021)** to remove old polluting vehicles; **ethanol blending**; **PM Ujjwala Yojana (2016)** LPG connections to cut indoor smoke.
- **Stubble and rivers:** the **Crop Residue Management** scheme (machines such as the **Happy Seeder**), the **Pusa bio-decomposer** that rots straw in the field, and using straw for **2G ethanol and biomass power**; **Namami Gange (2014)** with sewage treatment plants and real-time monitoring of industrial effluents.
- **AP:** the **AP Pollution Control Board** grants consents and monitors industries along the coast (**Visakhapatnam, Kakinada, Nellore**); **Visakhapatnam, Vijayawada, Guntur, Nellore** and other cities are under **NCAP**; after the **LG Polymers leak (2020)**, rules on hazardous chemical storage near homes were tightened.

EXAM TRAP

Water Act 1974 (CPCB, SPCBs; consent to establish and operate; **2024 amendment**); **Air Act 1981** (after **Stockholm 1972**; control areas); **EPA 1986** (after Bhopal; umbrella); **EIA 1994/2006** (Category **A** Centre, **B** State; **public hearing**); **CRZ 1991, 2011, 2019**; **Jan Vishwas 2023** (penalties instead of jail); **Red, Orange, Green, White** (2016); continuous monitoring; **NGT 2010** (Delhi + Bhopal, Pune, Kolkata, Chennai; retired SC judge or HC CJ; **6 months**; appeals to SC); **absolute liability: Oleum 1986**; **Art 21: Subhash Kumar 1991**; **polluter pays, precautionary: Vellore 1996**; **public trust: Kamal Nath 1997**; **Vienna 1985, Montreal 1987 (CFCs)**; **Kigali 2016 (HFCs; India freeze 2028, cut from 2032)**; **Cooling Action Plan 2019**; **NCAP 2019, PRANA**; **Swachh Vayu Survekshan 2023**; **CAQM 2021, GRAP**; **Surat ETS 2019**; **BS-VI 2020**; **FAME I 2015, II 2019, PM E-DRIVE 2024, PM-eBus Sewa 2023**; **Scrappage 2021**; **Ujjwala 2016**; Happy Seeder, **Pusa decomposer**; **Namami Gange 2014**; AP: **APPCB, NCAP cities, LG Polymers lessons**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. The Central Pollution Control Board was set up under the Water Act, 1974.

2. The Air Act was passed in 1981.

3. The Environment (Protection) Act, 1986 was passed after the Bhopal gas tragedy.

4. The National Green Tribunal can be headed only by a serving judge of the Supreme Court.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: C. The NGT is chaired by a retired Supreme Court judge or a retired High Court Chief Justice.

PRACTICE

Match List-I (principle) with List-II (case):

List-I (principle)	List-II (case)
A. Absolute liability	1. Subhash Kumar v. State of Bihar (1991)
B. Polluter pays and precautionary principle	2. M.C. Mehta v. Kamal Nath (1997)
C. Public trust doctrine	3. Vellore Citizens' Welfare Forum (1996)
D. Clean environment as part of Article 21	4. M.C. Mehta, Oleum gas leak (1986)

(1) A → 4 B → 3 C → 2 D → 1

(2) A → 3 B → 4 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (1) Oleum: absolute liability; Vellore: polluter pays and precaution; Kamal Nath: public trust; Subhash Kumar: Art 21.

PRACTICE

Which one of the following pairs is NOT correctly matched?

A. Montreal Protocol : ozone-depleting substances

B. CRZ Notification : air quality in cities

C. EIA Notification : environmental clearance for projects

D. Kigali Amendment : phase-down of HFCs

Answer: B. The CRZ Notification regulates activities along the coast, not urban air quality.

PRACTICE

Assertion (A): The Kigali Amendment aims mainly to phase out CFCs in order to protect the ozone layer.
Reason (R): HFCs do not deplete the ozone layer but are powerful greenhouse gases.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: D. A is false: Kigali targets HFCs (CFCs were handled by the original Montreal Protocol). R is true.

PRACTICE

A company plans a large new cement plant. Before it can get approval, it must study the likely effects on air, water and people and hold a public hearing. Under which rule does this happen?

A. National Clean Air Programme

B. Environmental Impact Assessment Notification

C. Graded Response Action Plan

D. Coastal Regulation Zone Notification

Answer: B. The EIA Notification (2006) under the EPA requires impact studies and public hearings before environmental clearance.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the laws in time order with one feature each, plus EIA, CRZ and industry categories. 2. Write the NGT facts, the four court principles with cases, and Montreal and Kigali. 3. Write the recent initiatives for air, transport, stubble and rivers, and AP's role.

NOTEBOOK PAGE · Day 31: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

GEOGRAPHY: Physiographic divisions of India

Area	Write

India's size and location	Area, rank, extreme latitudes and longitudes, N–S and E–W lengths, borders, coastline, standard meridian
Six divisions	Himalayas (ranges, heights, passes, regional divisions, Purvanchal); plains (belts, doabs, Majuli); plateau (highlands, Ghats, peaks, Meghalaya); desert; coasts (names, lagoons); islands (channels, volcanoes); AP's relief

ECONOMY: 1991 reforms

- Causes of the crisis with figures; leaders and dates; stabilisation vs structural adjustment; LPG measures (licensing, public sector, MRTP, FERA, devaluation, convertibility, tariffs, FDI, WTO); committees; disinvestment and DIPAM; results and criticisms; second-generation reforms.

SCIENCE: Pollution control

Write: laws with years and one feature each; EIA, CRZ, industry colour categories; NGT; court principles with cases; Montreal and Kigali; recent initiatives (NCAP, PRANA, Swachh Vayu Survekshan, CAQM, FAME, PM E-DRIVE, scrappage, Ujjwala, stubble steps, Surat emissions trading); AP's role.